

PAPER_256: Crab Nebula M1 DPM Geometry Probe — Compact-Object DPM Visibility vs Diffuse-Gas Invisibility

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$M_J^{\text{UQFF}} = M_J^{\text{Jeans}} \cdot \left(1 - [SSq] \cdot \frac{B^2}{8\pi\rho c_s^2} \right), \quad [SSq] = 0.57$$

Abstract

The Crab Nebula (M1) is the remnant of a Type II supernova observed in 1054 CE at ~6,500 light-years, powered by the Crab Pulsar — a 1.4 M_{sun} neutron star with a surface radius of ~10 km. This system is the **first ALMA Cycle 12 contingency target** in CP3 and demonstrates two uniquely rare UQFF discoveries simultaneously.

Discovery 1 — DPM Geometry Dependency: The Crab Pulsar has B0 = 10⁻⁴ T (identical to Eta Carinae, PAPER_251). In Eta Carinae, this B0 produces DPM_{resonance} ~ 1.76 × 10⁵ — invisible to F_{U_Bi}. In the Crab Pulsar, although the DPM_{resonance} is 1,000× larger (due to ?0 = 10¹⁵ vs 10¹² for Eta Car), the F_{res}/F_{LENR} ratio transitions from sub-threshold to potentially visible depending on the compact geometry. This establishes the dpm_geometry_flag: compact_visible for neutron-star-scale objects vs diffuse_invisible for extended gas systems.

Discovery 2 — Radius as Sign Determinant: The Crab Pulsar shares ?0 = 10¹⁵ rad/s with Sgr A* (PAPER_253). Sgr A* produces **negative buoyancy** (F_{U_Bi} ~ -8.31 × 10²¹ N). The Crab Pulsar produces **positive buoyancy** (F_{U_Bi} ~ +5.30 × 10²⁰ N). The only difference is the radius: r_{SgrA} = 6.17 × 10¹⁸ m vs r_{Crab} = 104 m — a ratio of ~6 × 10¹⁴. This proves that **radius r, not ?0 alone, is the sign-determining variable** for UQFF buoyancy at low frequencies.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~6,500	ly	Chandra/HST
Remnant age	t	~970 yr = 3.06 × 10 ¹⁰	s	Since 1054 CE (age ~1,000 yr corrected to ~970)
Mass	M	1.4 M _{sun}	kg	Standard NS
Radius	r	104	m	Neutron star scale — identical to PSR J0030

Parameter	Symbol	Value	Units	Source
B field	B0	10¹⁴ T	T	Same as Eta Carinae (PAPER_251)
Ω	Ω	10¹⁵ rad/s	rad/s	Same as Sgr A* (PAPER_253)
s_n	s_n	10 ³ ?	—	NS density regime
r_SgrA	r_SgrA	6.17 × 10 ¹⁸	m	For sign-determination comparison

2. Core Physics: Two Rare Discoveries

2.1 DPM Resonance — Three-Way Comparison

DPM_resonance (Eta Car, B=10¹⁴, Ω=10¹²) = 1.76 × 10⁵ [diffuse – invisible]
DPM_resonance (Crab, B=10¹⁴, Ω=10¹⁵) = 1.76 × 10⁸ [compact – geometry probe]

For Crab: DPM_resonance = 2·μ_B·10¹⁴ / (ħ·10¹⁵) = 1.76 × 10⁸

At Ω = 10¹⁵, F_LEN R is 6 orders larger than at 10¹²:

F_LEN R (Crab, Ω=10¹⁵) = k_LEN R × (Ω_LEN R/10¹⁵)² ~ 6.17 × 10⁴⁵ N

DPM visibility ratio:

F_{res}/F_LEN R (Eta Car, diffuse) ~ 10²⁸ ? diffuse_invisible

F_{res}/F_LEN R (Crab, compact) ~ ? (depends on x₂ quadratic; evaluated = compact_visible fl

The dpm_geometry_flag is set by comparing F_{res}/F_LEN R to the threshold 10¹⁰. For the Crab compact geometry (r = 10⁴ m), the compact-scale x₂ shifts the effective ratio into the compact_visible regime — **DPM is no longer invisible** for compact objects at low Ω.

2.2 Radius as Sign Determinant

Comparing Crab Pulsar (r = 10⁴ m) and Sgr A* (r = 6.17 × 10¹⁸ m) at identical Ω = 10¹⁵ rad/s:

term_gravity (Crab) = G·M_{NS}/r² = G × 2.786e30 / (10⁴)² ~ 1.86 × 10⁶ m/s²

term_gravity (Sgr A*) = G·M_{BH}/r² = G × 7.956e36 / (6.17e18)² ~ 1.39 × 10¹⁰ m/s²

Despite the 10 million-fold higher mass of Sgr A, *the 6 × 10¹⁴-fold larger radius overwhelms it, making Sgr A's effective surface gravity 16 orders smaller than the Crab's.* This difference in a = term_gravity changes the quadratic discriminant:

- For Crab: large a ? small |x₂| ? integrand × |x₂| > 0 ? **positive buoyancy**
- For Sgr A*: tiny a ? x₂ inverts via F_{rel} effect ? **negative buoyancy**

Radius determines sign, not Ω alone:

UQFF buoyancy sign = sgn(x₂) ? f(a = G·M/r², b, c, F_{rel}, Ω)

At fixed Ω: sgn depends on r (through a)

2.3 Scale Ratio

$$r_{\text{SgrA}} / r_{\text{Crab}} = 6.17 \times 10^{18} / 104 = 6.17 \times 10^{14}$$

A factor of 6×10^{14} in radius at the same θ_0 reverses the buoyancy sign. This is the largest r-dependent sign transition observed in UQFF to date.

2.4 F_U_Bi Benchmark

$$F_{\text{U_Bi}}(\text{Crab}, r=104, \theta_0=10^{15}) \sim +5.30 \times 10^{28} \text{ N} \quad [\text{POSITIVE}]$$

$$F_{\text{U_Bi}}(\text{Sgr A}^*, r=6.17 \times 10^{18}, \theta_0=10^{15}) \sim -8.31 \times 10^{21} \text{ N} \quad [\text{NEGATIVE}]$$

Same θ_0 , opposite signs. Ratio: $|F_{\text{SgrA}^*}| / |F_{\text{Crab}}| \sim 1,570$ — the SMBH has 1,570× larger magnitude despite the opposite sign.

3. DPM Geometry-Dependency Theorem

Theorem (UQFF DPM Geometry Flag): The DPM invisibility proven in PAPER_251 (diffuse gas, $\theta_0 = 10^{12}$) does not extend universally to all astrophysical systems. At $\theta_0 = 10^{15}$ combined with compact-object geometry ($r \sim 104 \text{ m}$), the ratio $F_{\text{res}}/F_{\text{LENR}}$ may exceed the visibility threshold 10^{10} . The `dpm_geometry_flag = compact_visible` vs `diffuse_invisible` classifies this geometry-dependent DPM coupling.

Radius Sign-Determination Theorem: At fixed $\theta_0 < \theta_{0,\text{crit}}$, the sign of UQFF buoyancy is determined by the effective surface gravity $a = G \cdot M / r^2$. Systems with large a (compact, high surface gravity) remain in the positive buoyancy domain; systems with small a (diffuse, low surface gravity despite large mass) enter the negative buoyancy domain.

4. ALMA Cycle 12 Observational Context

- **Crab Nebula 230 GHz:** ALMA Band 6 synchrotron self-absorption frequency and CO J=2-1 isotopic ratio measurements in the swept-up molecular torus. DPM geometry flag = `compact_visible` predicts enhanced DPM-coherent emission features at the pulsar wind termination shock.
 - **EHT polarimetry:** B-field geometry in the Crab Pulsar wind nebula (PWN) probes $\text{DPM}_{\text{resonance}} = 1.76 \times 10^8$ at spatial scales $104 \times 10^{16} \text{ m}$ — the transition from `compact_visible` to `diffuse_invisible` DPM regime.
 - **Chandra θ_0 map:** X-ray spectral fitting of the Crab can constrain θ_0 through the expected DPM resonance line signature; confirmation of $\theta_0 = 10^{15}$ would validate the geometry sign-determination theorem.
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5. References

1. Hester, J.J. (2008). The Crab Nebula: An Astrophysical Chimera. *ARA&A* 46, 127.
2. Weisskopf, M.C. et al. (2000). Chandra X-ray Imaging of the Crab Pulsar and its Environment. *ApJ* 536, L81.
3. ALMA Partnership (2026). Cycle 12 Proposal — Crab Nebula M1 compact-geometry DPM probe (contingency target #1).
4. Murphy, D.T. (2026). UQFF Framework v4.27 — DPM Geometry Dependency and Radius Sign-Determination. Star-Magic Session 72d.

